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Zero-Shot-Learning for Plant Species Classification

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Abstract

Zero-shot learning (ZSL) is a machine learning technique that identifies the target classes without any observed data, using semantic information from some source classes as the basis of knowledge transfer. ZSL has emerged as a new paradigm in machine learning to solve the constraints of classical supervised learning. This study explores applying ZSL techniques to a plant species dataset containing rich attribute descriptions for each class. Our study uses the multimodal capabilities of Contrastive Language–Image Pre-training to predict unseen plant species classes without needing labeled examples during training. Our objective is to contribute to advancing biodiversity conservation efforts and promoting environmental sustainability by facilitating the automated recognition of plant species based on semantic attributes.

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1. Introduction

The field of object classification has garnered significant attention within the realm of computer vision due to its pivotal role in facilitating visual comprehension. In recent times, the domain of supervised learning has experienced significant advancements, with a particular emphasis on deep learning methodologies. The observed progress can be attributed to developing extensive, annotated datasets, using rapid processing platforms, and implementing effective optimization strategies.

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1.1. Brief overview of traditional supervised learning

In machine learning, traditional supervised learning involves training a model on a labeled dataset of input-output pairs. The objective is for the model to acquire the ability to map inputs to matching outputs so that it can accurately predict new, unobserved data. Using this method, the model is trained to minimize the difference between its predictions and the actual outputs. The training dataset is meticulously labeled, and each input is associated with the appropriate output. In traditional supervised learning, the fundamental concept revolves around labeled data, where each input sample in the training dataset is paired with a corresponding valid output label [1]. The model's primary objective is to achieve accurate predictions on new, unseen data by generalizing from the labeled instances within the training set. Common algorithms employed in this paradigm include linear regression for predicting continuous outputs, logistic regression for binary classification tasks, Support Vector Machines (SVM) applicable to regression and classification problems, and neural networks capable of handling intricate, non-linear mappings through multi-layer perceptrons. The instructional procedure involves defining a loss function as an indicator of the variance between actual and expected results, followed by an optimization process wherein the model's parameters are adjusted to minimize the loss function [2]. Testing and validation procedures ensure the model's generalizability by assessing its performance on diverse datasets.

Traditional supervised learning shows its adaptability to several domains in real-world applications. One widespread use is image classification, which uses the model's capability to recognize certain objects or patterns in pictures to enhance computer vision and related disciplines. Speech recognition is another important area of technology [2]. It is particularly good at turning spoken words into text, which makes voice-controlled interfaces and transcription services more effective. Furthermore, in Natural Language Processing (NLP), where tasks like machine translation, named entity identification, and sentiment analysis are performed, classical supervised learning is essential [3]. These examples demonstrate how flexible and broadly applicable supervised learning approaches are for solving challenging issues in speech recognition, picture analysis, and language tasks [4].

1.2. Limitations of traditional supervised learning

While robust in scenarios with labeled training data, traditional supervised learning faces notable limitations when confronted with unseen classes. One primary challenge lies in the assumption that the model has been exposed to representative examples of all possible classes during training, making it ill-equipped to handle instances from entirely new or unforeseen categories [5]. The inability to generalize to unseen classes can result in poor performance, as the model needs to gain the necessary knowledge to make accurate predictions for novel data [6]. Moreover, the concept of unseen classes extends beyond just the absence of labeled samples; it encompasses the difficulty of adapting to dynamic environments where new classes continually emerge.

1.3. Definition and significance of ZSL

Humans can distinguish an estimated 30,000 readily discriminable objects. In Irving Biederman's seminal paper titled "Recognition-by-Components: A Theory of Human Image Understanding," published in Psychological Review in April 1987, the author introduces the Recognition-by-Components (RBC) theory, a groundbreaking framework for understanding how humans visually recognize and interpret complex visual scenes [3]. According to Biederman, "geons" are basic, volumetric primitives that are used as the foundation for identifying more complicated things. Geons are proposed to be simple geometric forms, such as spheres, cones, and cylinders, that may be assembled and rearranged to create a vast array of other items. He introduces the concept of a parse tree—a hierarchical structure of geons representing an object—. This hierarchical representation makes it possible to identify items by their structural links. [3]

On the other hand, most current computer vision approaches require hundreds of labeled examples for each class of objects to develop a recognition model. The field of machine learning or "learning to learn" research has drawn more attention inspired by the capacity of people to recognize patterns without the need for visual cues and the high expense of collecting and annotating training samples. These studies aim to aid future recognition tasks by intelligently using previously gained knowledge. One significant and current problem in this field is to develop recognition algorithms

that can identify new visual classes without the need for labeled training examples, i.e., accurately detecting the class without having previously observed it. This specific field of research is called zero-shot learning (ZSL). [4] [5]

1.4. Applications of ZSL

Zero-shot learning (ZSL) is highly relevant in real-world applications when unknown classes pose difficulties for typical supervised learning algorithms. ZSL becomes essential when acquiring labeled data for every conceivable class is impractical or expensive. A prominent use case is image recognition systems, where the capacity to recognize objects or entities that were not included in the training set is crucial for adjusting to changing surroundings. ZSL also plays a role in natural language processing; it allows textual material that was not explicitly included during the model's training phase to be classified.

2. Related Works

[14] introduce a remarkably simple yet effective zero-shot learning approach. They suggest a streamlined framework that treats the relationships between features, attributes, and classes as a two-layer linear network with fixed top-layer weights. [15] propose another approach, rooted in manifold learning, that aligns the semantic space with the model space. This approach allows accurate recognition of visual features. Authors achieve superior discriminative performance by introducing "phantom" object classes between these spaces, optimized from labeled data. This method was demonstrated on the ImageNet Fall 2011 dataset. However, learning succinct visual categories poses a real challenge due to the difficulty in acquiring training images even in the presence of attributes or textual descriptions: certain visual concepts, like complex interactions such as two people dancing, are not easily describable.

The authors in [16] suggest a zero-shot-learning approach that takes advantage of visual abstraction to tackle these concepts. Focusing on people and their interactions, the proposed method enables the provision of training data by manipulating abstract visual representations. The authors demonstrated the idea of human pose datasets and a new dataset featuring complex interactions. The proposed approach surpasses several baseline methods by explicitly learning to map between the abstract and real-world domains to enhance alignment across the two domains. To enhance ZSL's discriminative capability, [17] draws inspiration from human creativity in generating new art and modeling the visual learning process of unseen categories. The study establishes a connection between ZSL and human creativity, with a shared goal of recognizing or creating the unseen. Through empirical evaluation of different benchmarks such as CUB and NABirds datasets and attribute-based ZSL across additional datasets (AwA2, aPY, and SUN), the authors' approach outperforms state-of-the-art methods by several percentage points. However, zero-shot learning has yet to be applied to climate change-related datasets in any of these approaches. This paper's contribution to environmental studies lies in applying zero-shot learning techniques to plant species datasets, effectively recognizing various plant species. This contribution is of particular significance as it helps in conservation efforts and mitigating the effects of climate change. Recognizing plant species is extremely important for conservation efforts, as it enables a better understanding of ecosystem dynamics and helps efficiently allocate resources, including water. By accurately identifying plant species, conservationists and policymakers can implement targeted strategies to conserve water resources, ultimately contributing to sustainability and resilience in the face of climate change.

3. Fundamentals of Zero-Shot Learning

3.1. Definition and significance of ZSL

Zero-shot learning (ZSL) is a machine learning paradigm that deals with the challenges of traditional supervised learning when confronted with novel or previously unknown classes. ZSL allows models to predict classes that were not included in the training dataset, in contrast to traditional methods that call for explicit instances of every class during training. The fundamental concept is to learn a mapping between the characteristics of the input and a semantic space where embeddings or attributes represent classes. Using the common semantic information, the model uses this learned mapping during inference to forecast classes it has never seen before [7]. Stated differently, ZSL aims to close

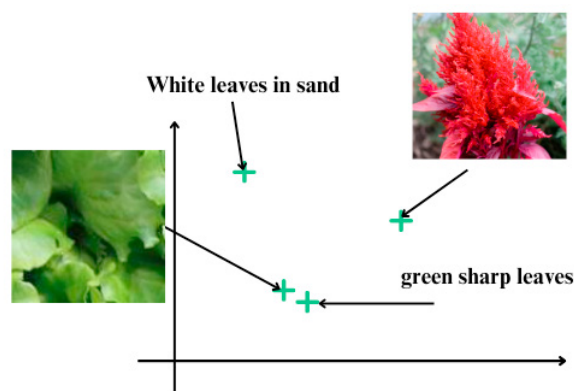


Fig. 1. Illustration of attribute-embedding. Text and images that are alike will be mapped to comparable vector spaces, whereas those that are dissimilar will not share similar vector spaces [23].

the semantic (and syntactic) [7] gap between the seen and unseen classes by employing information acquired from textual descriptions, such as word embeddings and attributes.

3.2. Explanation of the concept of zero-shot learning

Over the past ten years, significant advancements have been achieved in object detection in raw images. Reliable and effective detectors are available for certain object classes: faces, pedestrians, and vehicles. These detectors combine powerful low-level features like HoG [8] or SIFT [9] with cutting-edge machine learning techniques, like support vector machines (SVMs) [10], dimensionality reduction, outlier detection, and clustering [11], or boosting [12].

However, these systems need a large amount of manually labeled training data—many thousand sample photos are usually needed for each class that must be learned—to attain excellent classification accuracy [11]. Human learning functions differently: While it is true that humans learn and generalize effectively from examples, they may also recognize entirely new classes when given a high-level description. For instance, we can recognize a horse from the description “huge, tamed, plant-eating animal used for riding, racing, and hauling and pulling things. It has strong hooves and a flowing mane and tail,” and we can dependably identify mice from the description “a little rodent with a long tail, huge eyes and ears. Usually pointed in shape.” Building on this finding, the basic idea of ZSL is to categorize objects based on a set of high-level, semantically significant characteristics that we refer to as attributes.

3.3. ZSL vs Traditional Supervised Learning

Zero-shot learning (ZSL) models are trained on a labeled dataset containing instances of every class they may see during inference in classical supervised learning.[6] This restricts their capacity to extrapolate to novel or unobserved classes that aren’t included in the training set. Conversely, transfer learning entails pre-training a model on a source task and subsequently optimizing it on a target task. Models can use information from a related domain when fine-tuned through transfer learning, although examples from all target classes are still needed. On the other hand, ZSL goes in a completely another route by teaching models to identify classes they have never seen before [13]. The model may generalize to new classes based on common semantic information by learning a semantic space where attributes or embeddings represent classes. Thus, ZSL expands on traditional learning paradigms and provides a solution for situations when encountering classes that are not visible to oneself is a frequent and significant issue.

3.4. Challenges of ZSL

In order to effectively deploy zero-shot learning (ZSL), it is necessary to address several issues. One major challenge is the scarcity of data, which makes it impossible to collect labeled instances for all possible unknown classes.

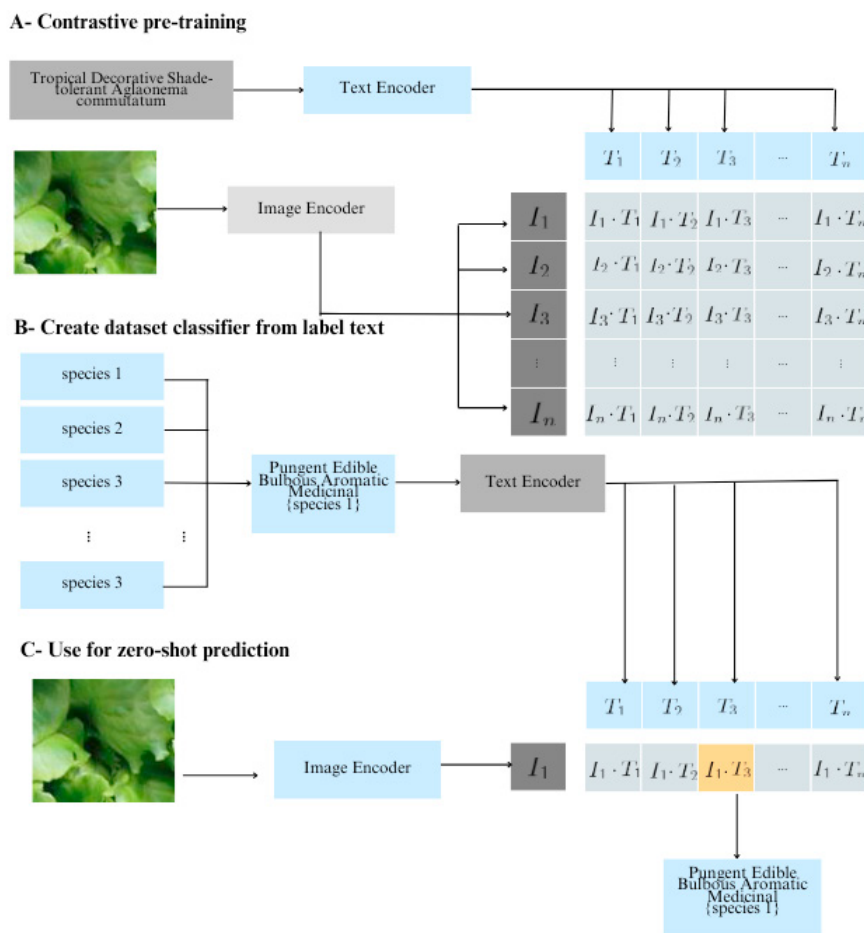


Fig. 2. Illustration of the principles of CLIP in ZSL. CLIP is initially trained using an image encoder and a text encoder to forecast images matching with corresponding texts in our dataset. Subsequently, we utilize this functionality to repurpose CLIP as a zero-shot classifier. This entails converting all class categories within a dataset into descriptive captions, like "an image showcasing a dog," and then predicting the class category of the caption that CLIP identifies as best aligned with a given image [21].

Insufficient training data for these unfamiliar classes could potentially impede the model's capacity to generalize effectively. Another challenge is the semantic gap, which refers to the disparity between the semantic representation of classes (such as attributes) and the plain feature space. Addressing this gap is crucial in order to attain consistent zero-shot performance. In addition, there are ongoing challenges in ensuring that the model can generalize to previously unknown classes and address biases in the training data. Another challenge that arises is the efficient scaling of ZSL models as the number of classes increases.

4. Methodology

4.1. Predictive Model

The modern world, which includes the digital and physical domains, is by its very nature multimodal, combining several senses—such as touch, smell, language, images, and video—into its operations. We see the world like an orchestra, combining these many inputs to create a cohesive internal picture, or "model," of our environment (see Fig 1). Both models effectively communicate by encoding similar concepts found in both text and images into comparable vectors. For instance, the text "two dogs running across a frosty field" would generate a vector closely resembling that

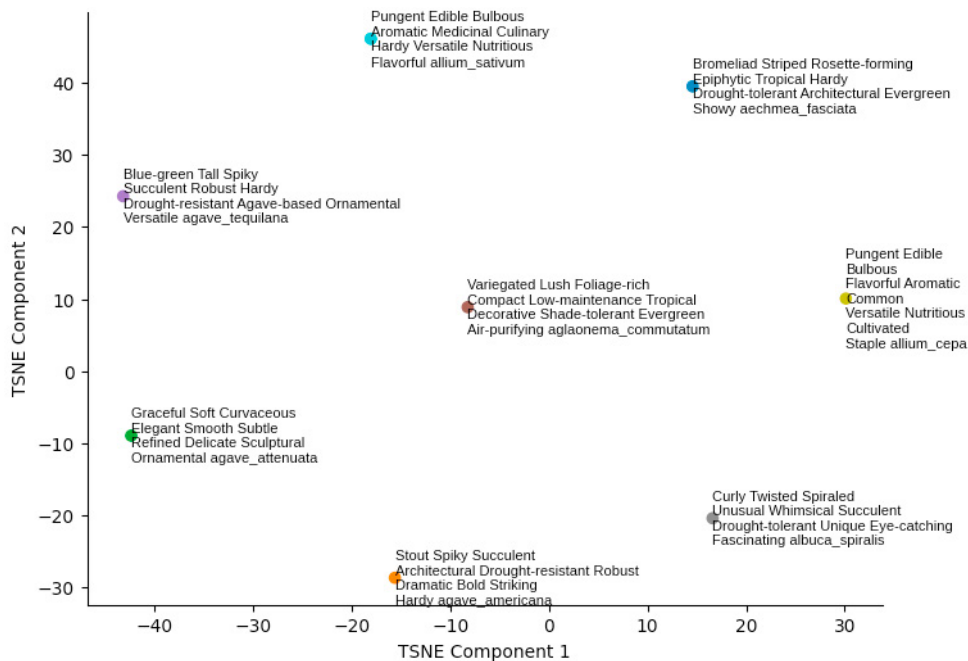


Fig. 3. Illustration of examples from the dataset.

of an image depicting two dogs running across a frosty field. In AI, multi-modality adoption is becoming increasingly important. Combining information from multiple modalities helps even specialized models only exclusively concerned with language or vision applications.

OpenAI introduced a neural network with the objective of tackling these challenges: it undergoes training using an extensive array of images alongside diverse natural language guidance sourced abundantly from the internet. The network's design allows it to receive instructions in natural language to undertake a wide range of classification tasks without the direct optimization for each benchmark's performance, akin to the "zero-shot" capabilities observed in GPT-2 [18] and GPT-3 [19].

CLIP (Contrastive Language–Image Pre-training) is built upon a substantial foundation of research concerning zero-shot transfer, natural language supervision, and multimodal learning. While the notion of zero-data learning has existed for over a decade, its exploration has primarily been within computer vision, exploring the capacity to generalize to unseen object categories [20].

Zero-shot performance can be achieved on various image classification datasets by scaling up a simple pre-training assignment. This uses a freely available source of supervision: text connected to photos collected from different websites. Using this data, a proxy training job for CLIP is developed, in which the model is asked to predict which of 32,768 randomly picked text snippets belongs to a particular image in our dataset. CLIP models develop the capability to recognize a broad spectrum of visual concepts depicted in images and associate them with their corresponding textual descriptions. Consequently, CLIP models become versatile and adaptable to nearly any visual classification task. For instance, in a dataset focused on distinguishing between photos of dogs and cats, a CLIP model is evaluated based on whether it is more inclined to predict the text description "a photo of a dog" or "a photo of a cat" for each image.

Real-world performance often falls short of the impressive benchmarks achieved by deep learning systems in vision tasks [22]. This disparity, known as the "benchmark-real performance gap," arises because these models tend to optimize solely for benchmark performance. In contrast, CLIP's evaluation of benchmarks doesn't involve direct training on their data, which helps eliminate the potential for such a mistake. As a result, CLIP's benchmark performance more accurately reflects its real-world capabilities. CLIP comprises two concurrently trained models: a 12-layer text transformer responsible for constructing text embeddings and either a ResNet or a vision transformer (ViT) responsible for

creating image embeddings. In language models, this involves capturing general rules and patterns in English, while in vision models, it entails capturing the characteristics of various scenes or objects.

Figure 3 illustrates the t-distributed Stochastic Neighbor Embedding (t-SNE) visualization of text embeddings generated by the Contrastive Language-Image Pre-training (CLIP) model. Each point in the scatter plot represents a text embedding corresponding to a specific label from our dataset. This two-dimensional representation allows for a comprehensive exploration of the semantic relationships between different text embeddings and their corresponding classes from the Dataset (See section 4.1).

The challenge with multimodality is that these models are typically trained separately and lack an inherent understanding of each other. CLIP addresses this issue through image-text contrastive pre-training, wherein text and image encoders are trained with consideration for the other modality and its context. This ensures that the text and image encoders develop a shared "indirect understanding" of patterns in both language and vision.

Contrastive pre-training operates by taking pairs of (text, image), where the text describes the image, and training the model to encode these pairs as closely as possible in vector space. The principle of Clip is as follows: Negative pairs can be directly derived from positive pairs. Given positive pairs (T_1, I_1) and (T_2, I_2) , we simply interchange the components to obtain negative pairs (T_1, I_2) and (T_2, I_1) . Subsequently, a loss function can be applied to maximize the similarity between (T_1, I_1) and (T_2, I_2) , while minimizing the similarity between (T_1, I_2) and (T_2, I_1) . Figure 2 illustrates a single pre-training step conducted on a single batch. The loss function is designed such that pairs along the diagonal are intended to have a maximized dot product score, while all other pairs should have a minimized dot product score. Both the text and image encoder models are optimized to adhere to this criterion [23].



Fig. 4. Illustration of examples from the dataset.

4.2. Dataset

The dataset we are using for this demonstration, titled "Plant Species" [24] consists of 800 entries, each corresponding to a distinct plant species. Figure 4 provides a visual representation of example images from different plant species. Each image is accompanied by its corresponding label. As illustrated in Table 1, the dataset contains a wide range of plant classes, with detailed descriptions provided for each species. These descriptions present essential traits and characteristics, ranging from physical attributes to ecological behaviors. These attributes and characteristics of plant species serve as essential components for generating image-text embeddings in zero-shot learning Contrastive Language-Image Pre-training. The detailed attributes provided in the textual descriptions can be used to create meaningful embeddings that capture the visual and semantic similarities between images and their corresponding plant species labels. By training CLIP on such data, the model learns to generalize its understanding across diverse plant species, enabling accurate recognition and classification of unseen plant images based on their textual descriptions. Using this dataset, we aim to study their unique features and ecological roles for further analysis and understanding. For instance, by accurately recognizing each plant species, we can significantly contribute to conserving water resources. Understanding the specific water needs and drought tolerance levels of various plant species allows for more

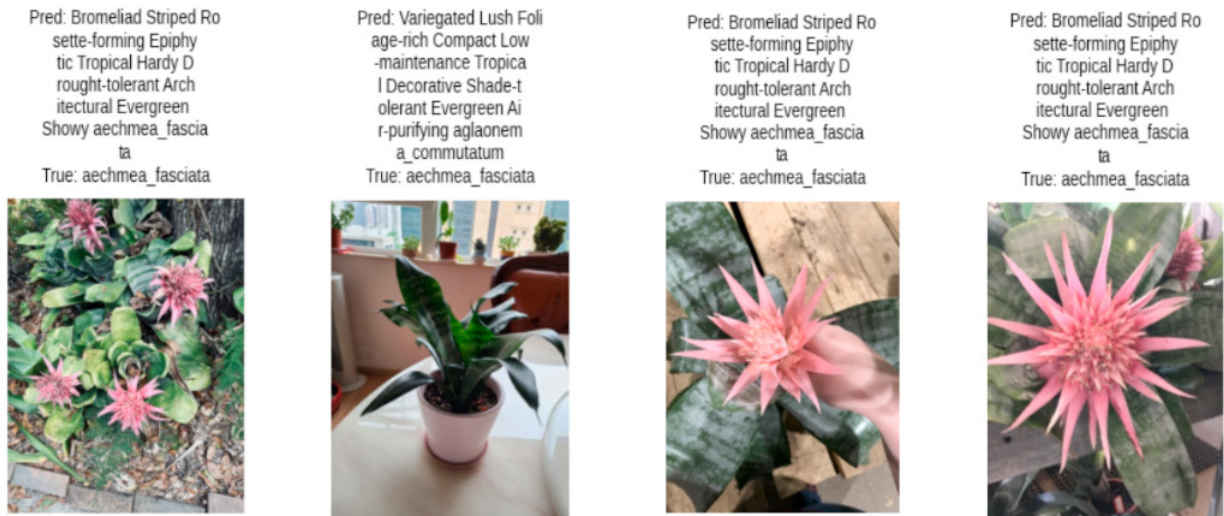


Fig. 5. Illustration of ground truth vs predicted labels

efficient agricultural practices. When we select plant species well-suited to local climate conditions and soil types, we can optimize water usage and reduce the need for excessive irrigation.

Table 1. An example of a table.

Class	Semantic Label (<i>t</i>)
aechmea_fasciata	Bromeliad Striped Rosette-forming Epiphytic Tropical Hardy Drought-tolerant Architectural Evergreen Showy
agave_americanus	Stout Spiky Succulent Architectural Drought-resistant Robust Dramatic Bold Striking Hardy
agave_attenuata	Graceful Soft Curvaceous Elegant Smooth Subtle Refined Delicate Sculptural Ornamental
agave_tequilana	Blue-green Tall Spiky Succulent Robust Hardy Drought-resistant Agave-based Ornamental Versatile
aglaonema_communitatum	Variegated Lush Foliage-rich Compact Low-maintenance Tropical Decorative Shade-tolerant Evergreen Air-purifying
albuca_spiralis	Curly Twisted Spiraled Unusual Whimsical Succulent Drought-tolerant Unique Eye-catching Fascinating
allium_cepae	Pungent Edible Bulbous Flavorful Aromatic Common Versatile Nutritious Cultivated Staple
allium_sativum	Pungent Edible Bulbous Aromatic Medicinal Culinary Hardy Versatile Nutritious Flavorful

4.3. Results

Our experimental results obtained a promising accuracy of 0.55875. Given that the pre-trained model has never seen the zero-shot classes before prediction. Figure 5 visually represents our model's performance using the plant species dataset. The Illustration displays sample photos with their true and predicted labels. Two instances are shown in each row; each indicates the accuracy of predictions. These results are promising and give us information on how well zero-shot learning methods can do to identify different plant species using semantic features without having previously been trained in said classes.

5. Conclusion

The intersection of ZSL and biodiversity science demonstrates considerable potential to tackle pressing issues such as climate change and the protection of natural ecosystems. Our methodology broadens the scope of ZSL's use beyond conventional domains and highlights its potential to support a range of practical uses, including conservation and

biodiversity monitoring. By applying semantic features for knowledge transfer, our work has shown the effectiveness of zero-shot learning (ZSL) in the task of classifying plant species.

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